

MSc Data Science Project

7PAM2002

Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Deep Learning for Diabetic Retinopathy Detection

Using Retinal Fundus Images

Student Name and SRN:

Saeed Murrad 23081869

Supervisor: Vidas Regelskis

Date Submitted: 21/04/2026

Word Count: 2510

DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at [Assessment Offences and Academic Misconduct](#) and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

I hereby give permission for the report to be made available on module websites provided the source is acknowledged.

Student Name printed: Saeed Murrad

Student Name signature: Saeed Murrad

Student SRN number: 23081869

UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

Abstract

This study discusses how deep learning can be applied in the automatic detection of diabetic retinopathy in retinal fundus images. Diabetic retinopathy is one of the leading causes of preventable blindness and timely diagnosis is essential in ensuring that the level of blindness is reduced. The paper was meant to address the limitations of manual screening by implementing a transfer learning method on three convolutional neural network architectures that are pretrained and use MobileNetV2, EfficientNetB0, and ResNet50. The five-class data were reduced to a binary classification problem, in which the images were classified into no DR, and DR Present. Image preprocessing included image resizing, data normalization, data augmentation, stratified dataset split, and upsampling to reduce the imbalance of classes. The models were trained and evaluated by the accuracy, precision, recall, F1-score, and confusion matrix analysis. The results showed that the three models were performed averagely and equally. The models had a range of accuracy ranging between 0.68 and 0.69. ResNet50 got the best overall performance, and its highest accuracy was approximately 0.69, precision of 0.44, recall of 0.56, and F1-score of 0.49. The lowest performance was that of EfficientNetB0 and moderate performance followed by MobileNetV2. The confusion matrix analysis also indicated that the ResNet50 detected the highest number of cases of diabetic retinopathy yet, false positives and false negatives were still present in all the models. Overall, the findings suggest that the transfer learning technique can be utilized to identify diabetic retinopathy and that ResNet50 is the most effective among all the analysed ones. However, the average results indicate that further optimization, increased scale training, and fine-tuning are required before these kinds of systems can be considered to be dependable in the real-life clinical practice.

Contents

Chapter 1: Introduction	1
1.1 Problem Statement	1
1.2 Research Aim	1
1.3 Research Objectives	2
1.4 Research Questions	2
Chapter 2: Literature Review	3
2.1 Related Work	3
Chapter 3: Dataset Description & EDA	5
3.1 EDA	5
Chapter 4: Ethical Implications	7
Chapter 5: Methodology	8
5.1 Dataset Description	8
5.2 Data Preprocessing	8
5.3 Handling Class Imbalance	8
5.4 Model Architecture	8
5.5 Model Training	9
5.6 Model Evaluation	10
5.7 Comparative Analysis	10
Chapter 6: Results	10
6.1 Performance Metrics Comparison	10
6.2 Accuracy	11
6.3 Precision	11
6.4 Recall	11
6.5 F1-Score	12
6.6 Confusion Matrix Analysis	12
6.7 MobileNetV2	12
6.8 EfficientNetB0	13
6.9 ResNet50	13
6.10 Summary	14
Chapter 7: Discussion	14
7.1 Overall Model Performance	14
7.2 Impact of Data Preprocessing and Class Balancing	14
7.3 Comparison with Existing Literature	15
7.4 Misclassified Cases and Possible Explanations	15
Chapter 8: Conclusion and Future Work	15
Github Repository Link	16
References	17

Chapter 1: Introduction

Diabetic retinopathy (DR) is an eye disorder that occurs progressively as the retinal blood vessels are damaged as a result of chronic diabetes (Alyoubi et al., 2020). It is also among the major causes of avoidable blindness in the world especially among adults of working age. The increasing cases of individuals at risk of developing diabetic retinopathy has been attributed by the World Health Organization as a result of the increasing prevalence of diabetes in the world today. The other important part of preventing the loss of vision is the early diagnosis of the disease as the condition can be slowed down or even prevented when timely treated (Tymchenko et al., 2020).

The diagnosis of DR occurs traditionally by manual examination of the images of the retinal fundus by ophthalmologists (Nguyen et al., 2020). This is effective, but time consuming, resource intensive and not applicable in low resource settings. The pressure of screening has put health care systems under a strain that the need to have an effective and scalable diagnostic solution has been devised (Doshi et al., n.d.).

New opportunities in automated analysis of medical images arose due to the current trends in Deep Learning (Mushtaq and Siddiqui, 2021). One specific type of deep learning algorithm is convolutional neural networks (CNNs) that can learn much more complex visual patterns, and have shown impressive performance on image classification tasks. The models have the potential to be trained to identify features that are linked to the disease in retinal fundus images with high precision in relation to diabetic retinopathy (Bhaskar, 2022).

The application of the concept of transfer learning is seen in the research report as the pretrained CNN models, including MobileNetV2, EfficientNetB0 and ResNet50 are applied to create an automated system that would be used to diagnose diabetic retinopathy. The problem will be easier to detect by simplifying the detection task in terms of binary classification problem (presence vs. absence of DR) and its reliable performance as well. These systems can help clinicians, enhance the effectiveness of screening, and early diagnosis, particularly in underserved areas (Sarki et al., 2020).

1.1 Problem Statement

The increasing number of diabetic patients has negatively affected the healthcare system despite the number of the screening methods, which are available. The diagnosis of diabetic retinopathy is done manually making it not only labour-intensive, but also subject to human error and inter-clinician variability. Moreover, in most areas, there is no sufficient number of skilled eye care specialists, and access to early diagnosis is restricted. This leads to the delay of treatment and permanent blindness. Thus, the urgent need is the development of an efficient, accurate, and scalable solution that will help to detect diabetic retinopathy in its early stages.

1.2 Research Aim

This research will focus on creating and testing an automated system to diagnose diabetic retinopathy based on retinal fundus imaging via a deep learning method. The research

aims at applying transfer learning methods based on the use of ready-trained convolutional neural network structures such as MobileNetV2, EfficientNetB0, and ResNet50 to provide binary classification of diabetic retinopathy (present or absent). It seeks to study the performance of these models when they are trained with a balanced set of data with the use of preprocessing and augmentation techniques.

1.3 Research Objectives

- To preprocess retinal fundus images, including resizing, normalization, and data augmentation
- To convert the original dataset labels into a binary classification problem (DR vs No DR)
- To split the dataset into training, validation, and test sets with stratification
- To address class imbalance in the training data through upsampling techniques
- To implement transfer learning using pretrained models: MobileNetV2, EfficientNetB0, and ResNet50
- To train the models using a TensorFlow data pipeline and evaluate their performance
- To compare model performance using metrics such as accuracy, precision, recall, F1-score.

1.4 Research Questions

- How effectively can deep learning models detect diabetic retinopathy from retinal fundus images?
- Does converting the problem into binary classification improve model performance and simplicity?
- Which pretrained model (MobileNetV2, EfficientNetB0, or ResNet50) performs best for this task?
- How does class balancing (upsampling) affect model performance?
- What is the impact of data preprocessing and augmentation on model accuracy and generalization?
- Which evaluation metrics best reflect the reliability of the models in detecting diabetic retinopathy?

Chapter 2: Literature Review

Diabetic retinopathy is the leading cause of blindness in the world and early and proper diagnosis is the greatest determinant in ensuring that the blindness is treated and prevented properly. The progress in deep learning over the last couple of years has contributed to the improved automated retinal fundus image analysis which has been the most promising to replace the traditional diagnostic methods. The literature review looks at a number of studies that analyse the application of deep learning techniques, in particular, convolutional neural networks, to detect and categorize diabetic retinopathy. The methods that have been considered in the reviewed articles are transfer learning, ensemble modelling and better neural network architecture that have been shown to improve diagnostic accuracy.

2.1 Related Work

It has been proposed that (Simul Hasan Talukder and Author, n.d.) use a more complex model of deep learning to identify diabetic retinopathy through the introduction of transfer learning and ensemble learning. The hypothesis of the experiment is that a team of pretrained convolutional neural networks such as ResNet50, DenseNet, MobileNet V2 and Inception models will be compared and the one which scores most will be picked. It is achieved through data augmentation and regularization to achieve more optimal results and minimize overfitting and maximize generalization. The authors also streamline the outcomes of weighted and average ensemble combination of a big number of models. They yield good results which are highly accurate and correct and their F1-scores and models are nearly flawless when it comes to classification. The usefulness of the implementation of the pretrained models and ensemble learning to increase the accuracy of the diagnostic is proven in this paper, yet the results would probably have to be verified on more heterogeneous data.

(Sarki et al., 2020 provide an extensive overview of deep learning pipelines that are employed to identify diabetic eye disease, including datasets, preprocessing, model architecture, and performance measures. Their discussion shows that deep learning can always be more accurate in the classification of data, than traditional methods, and that their major problems are the lack of annotated data, high-computational cost and low generalization. It also puts an emphasis on transfer learning as a special-use strategy in smaller datasets.

The review of CNN-based approaches to the detection and classification of DR and, in particular, the capacity of the models to extract features in a hierarchical order, without human intervention is presented in (Tsiknakis et al., 2021). The authors highlight the quality of data and class balance, and state that there remain issues with cross-population generalization and standardization, as well as should have more easily understandable models and clinical integration.

(Abushawish et al., n.d.) compare transfer learning, end-to-end learning and hybrid approaches on various datasets and discover the class imbalance and variations in data as

the most common problems. They promote explainable AI, like Grad-CAM, to enhance transparency and develop clinical trust and focus on models that can predictably generalise to new data sources.

The study (Rakib Ul Islam Rahat et al., 2025) contextualizes AI-based DR detection in clinical practice, noting the increasing disease burden across the world and requirements of scalable screening systems. They refer to new technologies such as portables (imaging equipment) and telemedicine systems, but they also cite regulatory and image quality issues as impediments to the practical implementation of these proposals.

Both (Bhulakshmi and Rajput, 2024) and (Nadeem et al., 2022) approve the statement that deep learning is better than traditional diagnostic systems in identifying lesions in the retina, and that the cost of annotations, the complexity of the calculation, and the interpretability of the model are typical problems. Both papers demand powerful, explicable models that are proven to be applicable in clinical practice.

Chapter 3: Dataset Description & EDA

The dataset in the present study will be acquired in Diabetic Retinopathy Unzipped Dataset, a cleaned and fixed version of original dataset that consists of high-resolution retina fundus images. Images that are corrupted are removed in this version which ensures that there is quality and consistency of the data in the model training process. The data set is composed of the retina images in different conditions of imaging that provided variation in the light, focus and orientation. The subjects have two pictures, indicating the right eye and the left eye, and a new subject ID (i.e., 1 right, 1 left). All the images are rated by the clinician on a scale of 0 through 4, with 0 signifying no diabetic retinopathy, and 4 signifying proliferative diabetic retinopathy. Multi-classification of this study is reduced to a binary classification where there is Level 0: No DR and Level 1-4: DR present. This eases the detection process whilst maintaining clinically significant information.

3.1 EDA

Figure 1 shows that the dataset is imbalanced, with 7,300 images labeled as No DR and 2,700 images labeled as DR Present (approximately 73% vs. 27%).

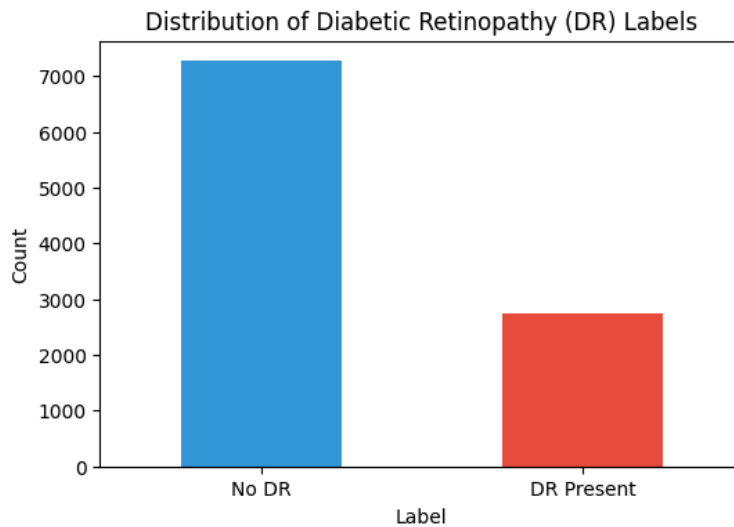


Figure 1 Distribution of Diabetic Retinopathy Labels

Figure 2 displayed retinal images are classified as “No DR,” showing consistent normal fundus patterns without visible signs of diabetic retinopathy.

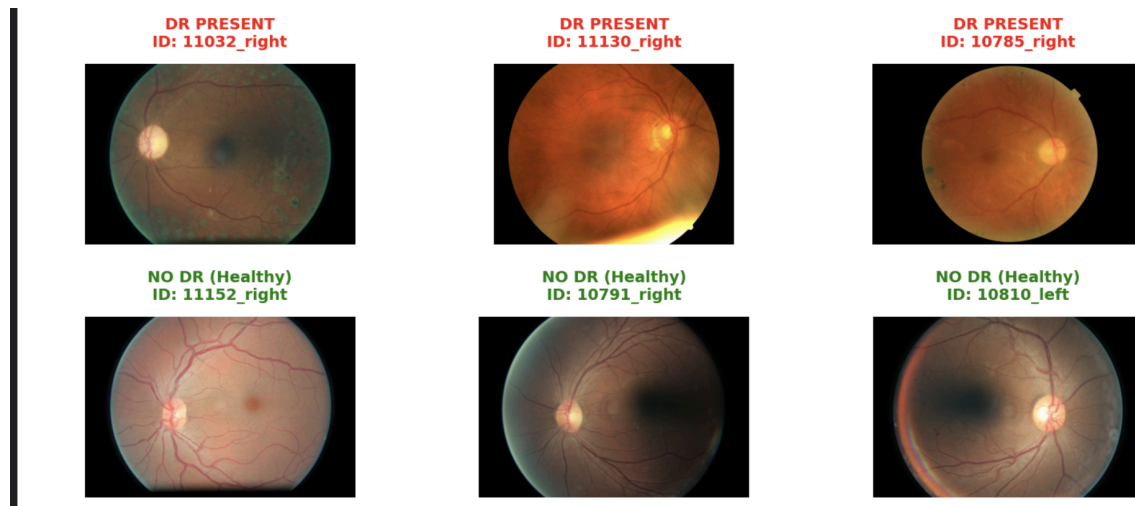


Figure 2 Labels

Figure 3 depicts that the dataset is skewed towards Level 0 (No DR) (7,300 images), where 750, 1,600, 250, and 200 images represent Levels 1-4 respectively. Here is a high level of imbalance of classes in terms of severity.

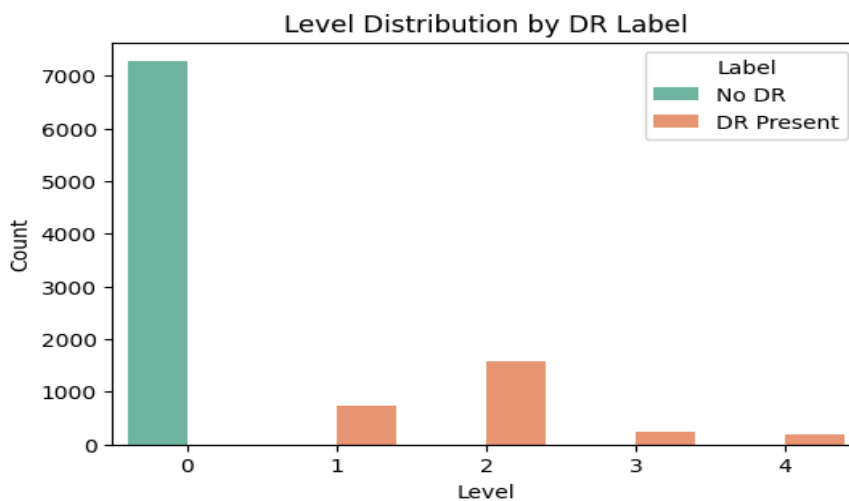


Figure 3 Level Distribution by DR Label

Chapter 4: Ethical Implications

This study has been conducted in line with ethical requirements and processes of conducting research in the University of Hertfordshire. The publicly available anonymised data was employed in the current study and no data of the patients will be identified. In such a way, the study will not presuppose any direct contact with human subjects or personal information of sensitive character, which will minimize the possibilities of any ethical damage. However, some ethical issues might arise when it comes to medical imaging data and artificial intelligence application. One of the most significant issues is the problem of privacy and confidentiality of the data, and there is no other solution to this question than the utilisation of the anonymised information of the credible sources. In addition, the bias in the data may also affect the impartiality and the external validity of the developed models. Having a poorly prediction of the model on underrepresented populations may be a risk in case the dataset does not cover many types of populations. This is a weakness that is achieved and we make sure that we do not misinterpret the results in an irresponsible way. The other ethical issue is the problem of the misuse or excessive use of automated systems in clinical decision-making. The created model will be implemented as a decision-support tool instead of a substitute of the medical professionals. Assessment and professional control: It must undergo clinical validation and professional control before it is able to be put into use. Besides, AI systems should be trustworthy and the model predictions must be transparent and accountable. Overall, the study will contribute to the further development of the concept of responsible deep learning in the healthcare environment without violating the ethical considerations and institutional provisions.

Chapter 5: Methodology

The chapter is an explanation of the strategy adopted to come up with a deep learning-based system so as to identify diabetic retinopathy in the support of retinal fundus images. It describes the used data, data preprocessing, model development, training and evaluation methodology. The approach is based on the principles of transfer learning, which presupposes the utilization of the already trained convolutional neural networks, which can extract features successfully and improve classification. There is a systematic pipeline based on the analysis which involves the preparation of data, training and performance evaluation of a model.

5.1 Dataset Description

The dataset used in this study consists of retinal fundus images of a publicly available diabetic retinopathy dataset. The labels, which describe the disease severity, are correlated to the pictures. To meet the purpose of this study; the multi-class labels were converted to binary classification problem with the images already classified as either No Diabetic Retinopathy or Diabetic Retinopathy Present. This simplification allows the model to focus on finding the existence of the disease, but not the severity. The computational efficiency was also improved by using a smaller sample of the dataset to avoid spending excessive time to train the model with an equal representation of the two classes.

5.2 Data Preprocessing

Before the input data can be used, it is important to process the data to assure quality and consistency. All retinal images were rescaled to a standard size of 224 x 224 pixels to meet the requirements of the input of pre-trained convolutional neural networks. The pixel values were scaled between 0 and 1 in order to enhance model convergence. Also, the images whose file paths were missing or invalid were filtered because it would create errors in the course of processing. In order to improve model generalization and minimize overfitting, the training dataset was subjected to data augmentation (random flipping, rotation, zooming). These variations are modelled to represent variations in retinal images in the real world.

5.3 Handling Class Imbalance

Class imbalance is also characteristic of medical datasets, in which a type of data is well represented by the other. The imbalance was minimized in this study by using an upsampling method on the minority group in the training set. In so doing, the samples in the minority group are multiplied by expanding on the already existing samples, therefore the model is not biased to the majority group. It is important to note that balancing method was applied on the training set only to maintain the integrity of validation and test data.

5.4 Model Architecture

Three pretrained convolutional neural network models, including MobileNetV2, EfficientNetB0, and ResNet50, are applied in this study to transfer learning. These

models were selected due to the fact that they are efficient in the classification of images and the fact that they can extract high level features on images. Weights that were trained on ImageNet were used to initialize the trained models. These models were custom classified by having a global pooling layer followed by a dense layer with sigmoid activation function which was used in binary classification. The lower levels of the models were frozen to prevent the loss of characteristics previously learned as well as to reduce the amount of computation power. Table below summarises their key architectural properties:

Property	MobileNetV2	EfficientNetB0	ResNet50
Total parameters	~3.4M	~5.3M	~25.6M
Depth (layers)	53	82	50
Input size	224×224	224×224	224×224
Core building block	Inverted residual + depthwise separable conv	Mobile inverted bottleneck (MBConv)	Residual bottleneck block
Activation function	ReLU6	Swish (SiLU)	ReLU
Normalisation	Batch normalisation	Batch normalisation	Batch normalisation
Pooling	Depthwise separable (stride 2)	Squeeze-and-Excitation + stride conv	Max pooling (stride 2)
Classifier head (custom)	GlobalAveragePooling2D → Dense(1, sigmoid)	GlobalAveragePooling2D → Dense(1, sigmoid)	GlobalAveragePooling2D → Dense(1, sigmoid)
Pretrained on	ImageNet	ImageNet	ImageNet
Base layers frozen	Yes	Yes	Yes

Table 1: Model Architecture for MobileNetV2, EfficientNetB0, and ResNet50

5.5 Model Training

The models were trained using the TensorFlow and Keras models. The loss was a binary cross-entropy loss function as it can be applied in binary classification problems. Adam optimizer was used to update the model weights. The training was done over a fixed number of epochs with early stopping being used to avoid overfitting. Early stopping is used to check the loss of validation and terminates training when there is no improvement in a set number of epochs. The training also made use of the batch processing which included a TensorFlow data pipeline and enhanced the performance and efficiency.

5.6 Model Evaluation

To determine the performance of the trained models holistically, the performance had a number of measures. Such metrics are accuracy, precision, recall, F1-score and Area Under the Curve (AUC). The overall correctness of the model is measured by the accuracy of the model; the recall is known as the false positives and the precision as the false negatives respectively. Precision and recall are balanced in the F1-score and it is especially helpful with imbalanced data. Confusion matrices were also created to show performance of the classifications and the trends of incorrect classifications. The test data was used to evaluate the program in order to have unbiased results.

5.7 Comparative Analysis

To analyse how the three pretrained models performed, the performance of the models was compared. The training and testing of all models in similar conditions were fair. The results were contrasted against each other in terms of the evaluation metrics and the most suitable model was selected. The comparison offers a hint of the effectiveness of various architectures in the detection of diabetic retinopathy and makes it possible to select the most appropriate model that will be used in the current task.

Chapter 6: Results

The chapter is an experimental study to compare three pretrained deep learning models, i.e., MobileNetv2, EfficientNetB0, and ResNet50, in the recognition of diabetic retinopathy. Each of the models is compared on the basis of four key performance measures which are accuracy, precision, recall, and F1-score. Analysis of the classification behaviour and pattern of misclassification is also done using confusion matrix analysis. Together, these tests can illustrate whether each model is good or bad in decoding retinal images, and recommend which architecture would be best in the problem of classifying retinal images.

6.1 Performance Metrics Comparison

Table 1 points out the accuracy, precision, recall and F1-score of each model on the test data. All three architectures had a general accuracy of approximately 0.68-0.69 with generally similar outcomes. ResNet50 demonstrated good results across the metrics since it was the highest ranked, followed by EfficientNetB0 and MobileNetV2 was the lowest ranked across the metrics.

Model	Accuracy	Precision	Recall	F1-Score
MobileNetV2	0.68	0.39	0.42	0.41
EfficientNetB0	0.68	0.42	0.49	0.45
ResNet50	0.69	0.44	0.56	0.49

Table 2: Summary of evaluation metrics for MobileNetV2, EfficientNetB0, and ResNet50

6.2 Accuracy

The three models have scores which are clustered around the accuracy with a range of about 0.68 (MobileNetV2 and EfficientNetB0) up to 0.69 (ResNet50). Even though the differences are not significant, they indicate that the deeper architecture of ResNet50 can be used to extract features of retinal images a bit more reliably. The small gap also shows that transfer learning is an effective method to use in this classification task irrespective of the architecture selected. However, even the relatively small absolute accuracy values of all the three models indicate obvious possibilities of improvement by means of fine-tuning, longer training or better preprocessing.

6.3 Precision

The ratio of predicted positive cases which are actually positive is also a crucial parameter in medical diagnostics, and in this case, it is desirable to minimise false alarms. The highest level of precision was attained by ResNet50 at around 0.44, then EfficientNetB0 at around 0.42 and finally MobileNetV2 at about 0.39. The fact that ResNet50 has a small lead does not mean that false positives are not a concern since the middle-level precision scores of all three models point to it. The rate of unnecessary positive predictions would need further refinement, such as better preprocessing and fine-tuning of models.

6.4 Recall

Recall measures how well a model can correctly identify all true positives - of particular importance in clinical practice where false positives are life threatening. ResNet50 once again topped the list with a recall of about 0.56, followed by EfficientNetB0 with 0.49 and MobileNetV2 with 0.42. The greater disparity between recall and precision shows that the depth has a significant involvement in the identification of the less salient visual characteristics of diabetic retinopathy. However, even with the relative advantage of ResNet50, relatively poor recall rates of all the models show that a clinically significant proportion of true positive cases would still be missed without further optimisation, like a better data augmentation or targeted fine-tuning

6.5 F1-Score

The F1-score is an unbalanced measure of precision and recall, which can be used particularly well in a situation of class imbalance. The F1-score of ResNet50 was maximum and was approximately 0.49, then EfficientNetB0, and MobileNetV2 had F1-score of 0.45 and 0.41 respectively. These findings support the trend seen at the level of single metrics: ResNet50 shows a consistent level of performance in comparison with other two architectures, and all three models exhibit the level of diagnostic balance of meaningful but improvable quality. Additional optimisation (such as hyperparameter optimization and data refinement) would probably provide significant gains in F1-score of all architectures.

6.6 Confusion Matrix Analysis

The confusion matrices below indicate the raw numbers of classification according to each of the models, the true and false number of each of the categories of No DR and DR. These results will be complementary to the aggregate measures since they will indicate the concentration of the errors in each model.

6.7 MobileNetV2

	Predicted: No DR	Predicted: DR	Total
Actual: No DR	340	101	441
Actual: DR	93	66	159

Table 3: MobileNetV2 confusion matrix

MobileNetV2 properly identified 340 No DR and 66 DR cases. Nonetheless, it generated 101 false positives - healthy cases that it mistakenly identified as DR - and failed to identify 93 actual DR cases (false negatives). This trend shows that the model has a fair degree of accuracy in healthy retinal images but it is not sensitive enough to accurately detect diabetic retinopathy. Particularly of clinical concern is the false negativity rate, which is quite high, and implies that the false alarms are not likely to hurt the patient as much as missed diagnosis does. The training plan and proportions of classes could be enhanced.

6.8 EfficientNetB0

	Predicted: No DR	Predicted: DR	Total
Actual: No DR	334	107	441
Actual: DR	82	77	159

Table 4: EfficientNetB0 confusion matrix

EfficientNetB0 was able to accurately detect 334 No DR cases and 77 DR cases. It had a slightly higher true positive, and a slightly lower false negative, (82 vs. 93) which indicates slight improvement in sensitivity compared to MobileNetV2. Nonetheless, it also produced 107 false positives, which is slightly higher than MobileNetV2, which suggests the over-prediction of the presence of the disease. All in all, EfficientNetB0 has a rather more balanced classification profile, but optimisation work remains to be done to decrease misclassification and enhance diagnostic reliability.

6.9 ResNet50

	Predicted: No DR	Predicted: DR	Total
Actual: No DR	326	115	441
Actual: DR	71	88	159

Table 5: ResNet50 confusion matrix

ResNet50 correctly identified 326 No DR and 88 DR cases, with the highest number of true positives of the three models. It also had the lowest false negative rate of 71 which validates its better sensitivity to detect cases of true diabetic retinopathy. The trade-off is more false positives of 115, that is, it tends to more frequently mark out healthy patients to be investigated further. This trade-off is usually more desirable in a clinical setting than false negative diagnoses since false positives can be addressed by follow-up. On the whole, ResNet50 shows the best diagnostic performance of the three architectures, but further decreasing the false positive rate with further tuning would enhance its practical value.

6.10 Summary

In all metrics of evaluation, ResNet50 was the best model in accuracy (0.69), precision (0.44), recall (0.56), and F1-score (0.49). Its confusion matrix output also proves that it identifies the highest number of true positive with minimum false negative rate- a very important aspect in medical screening uses. EfficientNetB0 was placed in the middle, slightly better than MobileNetV2 on the majority of the metrics. All three models showed that transfer learning is a good baseline method to classify diabetic retinopathy but the moderate level of performance of all architectures indicates the necessity of further improvement. Future directions to consider are domain-specific fine-tuning, more sophisticated models of augmentation, and experimentation with ensemble or attention-based methods to boost sensitivity and the overall diagnostic accuracy.

Chapter 7: Discussion

This chapter explains how the results of the study were in reference to the research objectives and questions. The results of the three pretrained deep learning models, such as MobileNetV2, EfficientNetB0, and ResNet50 are discussed in terms of the evaluation metrics and confusion matrices. The findings are also discussed in the frame of the available literature to reveal the similarities, differences, and major findings.

7.1 Overall Model Performance

The findings prove that the three models performed moderately similar in terms of evaluation measures, which proved that transfer learning is an effective method of diabetic retinopathy detection. Their accuracy rates were approximately 0.68 and 0.69 and this indicated that both models could acquire the required characteristics of retinal pictures. The performance levels are however merely average and this implies that the models can be used, although they are not in a position to be applied to the clinical setting until they are enhanced. The ResNet50 model performed better in all measures such as accuracy, precision, recall and F1-score compared to the MobileNetV2 and EfficientNetB0 models. This means that finer retina fundus images would be best described by finer architectures. The results of the current study are consistent with those of the other sources that support the benefits of the more sophisticated convolutional neural networks in the medical image classification.

7.2 Impact of Data Preprocessing and Class Balancing

The instances of preprocessing methods were resizing, normalisation and data augmentation, which were used to enhance the overall performance of the models in terms of better generalisation. Besides, the issue of imbalance of classes was also addressed with the help of upsampling that was employed to remove the bias towards the majority class. Nevertheless, the data set was still troublesome, which was also reflected in the intermediate assessment scores. It means that preprocessing and balancing are good but they cannot be applied either separately to achieve high-performance result.

Additional techniques to optimize the models are class weighting or artificial generation of data.

7.3 Comparison with Existing Literature

Its findings are in agreement with the previous research, which highlights the utility of the transfer learning to identify diabetic retinopathy. This study, as in those of Talukder et al. and Nadeem et al., indicates that pretrained models are capable of making reasonable performance with small amounts of data. However, the results obtained in this research are moderate compared to certain researches that have such large percentage of accuracy. This might be due to the disparity in the size of the dataset, the way of preprocessing or the evaluation strategies. It also highlights the point that realistic evaluation is critical and deep learning models are hard to apply on real world medical data.

7.4 Misclassified Cases and Possible Explanations

The analysis of the confusion matrices has shown that there are similar patterns of misclassification in all three models. False negatives -true DR cases were classified as No DR and false positives were classified as true DR - MobileNetV2 had the highest number of false negatives, 93 and ResNet50 had the lowest number of false negatives, 71. The errors are probably systematic features of the underlying images. Cases of DR that were initially characterized as Level 1 (graded as Level 1) are the most frequently misdiagnosed because some of the characteristics such as microaneurysms and small haemorrhages are minute and can resemble normal retinal variation especially when the images are poorly focused or have uneven lighting. False positives, in turn, are more likely to be produced by healthy images with artefacts or anatomical variations of features, which might be surface-level similarity to pathological lesions like prominent branching of vessels. The fact that ResNet50 trades fewer false negatives at the cost of more false positives indicates that it has a less robust architecture that can be more sensitive to subtle features, though at times it over-interprets normal variation as pathology. The systematic review of misclassified samples and Grad-CAM activation maps in the future would help to understand whether the errors are concentrated around certain image quality or severity levels, which would directly translate into specific preprocessing improvements.

Chapter 8: Conclusion and Future Work

This study has developed and tested a deep learning-based approach to the automatic detection of diabetic retinopathy by analysing retina fundus images. The research made the diagnostic intent simple by reducing the five-class grading issue into a binary classification issue such that the definition of whether or not diabetic retinopathy is present is established. Three models of the convolutional neural network, MobileNetV2, EfficientNetB0, and ResNet50, that have been trained were used on the dataset using the transfer learning and the accuracy, precision, recall, F1-score, and confusion matrices were used to assess the models. The findings revealed that all three of the models were

able to learn significant retinal features and perform the task of classification with moderate success. The most successful of them was ResNet50, i.e., it was the most accurate, precise, recalled, and with the best F1-score, and was also more efficient in detecting cases of diabetic retinopathy.

The next highly obvious observation in the study was that the preprocessing techniques such as resizing, normalisation, data augmentation and class balancing were positively correlated with model training and generalisation. However, the results were moderate overall, and it implies that the developed models are yet to achieve the high diagnostic performance level. All the models, in particular, had also false positives and false negatives, indicating shortcomings in the sensitivity and predictive power. Therefore, despite the fact that the study has shown that transfer learning can be helpful in predicting diabetic retinopathy, it also suggests that optimization is yet to be done before such systems can be applied in practice into the real world.

The experiments in the future should be focused on improving the performance of the models by fine-tuning the backbones rather than making them frozen. It may also be enhanced by training on more data, more training epochs and more sophisticated augmentation techniques to render it robust. In addition, further research may also focus on the multi-class classification to establish the level of diabetic retinopathy severity as opposed to binary identification. The proposed system would be more useful and reliable with the help of ensemble learning, explainable AI, such as Grad-CAM, and external clinical validation of the system on different datasets.

Github Repository Link

<https://github.com/saeedmurrad/fyp-ds-uoh>

References

- Abushawish, I. Y., Modak, S., Abdel-Raheem, E., Mahmoud, S. A., & Hussain, A. (n.d.). *Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000. Deep Learning in Automatic Diabetic Retinopathy Detection and Grading Systems: A Comprehensive Survey and Comparison of Methods*. <https://doi.org/10.1109/ACCESS.2017.DOI>
- Alyoubi, W. L., Shalash, W. M., & Abulkhair, M. F. (2020). Diabetic retinopathy detection through deep learning techniques: A review. *Informatics in Medicine Unlocked*, 20, 100377. <https://doi.org/https://doi.org/10.1016/j.imu.2020.100377>
- Bhaskar, D. N. (2022). *Diabetic Retinopathy Detection Using Deep Learning*. Springer. https://doi.org/10.1007/978-981-19-1559-8_39
- Bhulakshmi, D., & Rajput, D. S. (2024). A systematic review on diabetic retinopathy detection and classification based on deep learning techniques using fundus images. *PeerJ Computer Science*, 10. <https://doi.org/10.7717/PEERJ-CS.1947>
- Doshi, D., Shenoy, A., Sidhpura, D., & Gharpure, P. (n.d.). *Diabetic Retinopathy Detection using Deep Convolutional Neural Networks*.
- Mushtaq, G., & Siddiqui, F. (2021). Detection of diabetic retinopathy using deep learning methodology. *IOP Conference Series: Materials Science and Engineering*, 1070(1), 012049. <https://doi.org/10.1088/1757-899x/1070/1/012049>
- Nadeem, M. W., Goh, H. G., Hussain, M., Liew, S. Y., Andonovic, I., & Khan, M. A. (2022). Deep Learning for Diabetic Retinopathy Analysis: A Review, Research Challenges, and Future Directions. In *Sensors* (Vol. 22, Number 18). MDPI. <https://doi.org/10.3390/s22186780>
- Nguyen, Q. H., Muthuraman, R., Singh, L., Sen, G., Tran, A. C., Nguyen, B. P., & Chua, M. (2020). Diabetic retinopathy detection using deep learning. *ACM International Conference Proceeding Series*, 103–107. <https://doi.org/10.1145/3380688.3380709>
- Rakib Ul Islam Rahat, S., Rahman, H., Arafat, Y., Rahaman, M., Minzamal Hasan, M., & Al Amin, M. (2025). *British Journal of Nursing Studies Advancing Diabetic Retinopathy Detection with AI and Deep Learning: Opportunities, Limitations, and Clinical Barriers*. <https://doi.org/10.32996/bjns>
- Sarki, R., Ahmed, K., Wang, H., & Zhang, Y. (2020). Automatic Detection of Diabetic Eye Disease through Deep Learning Using Fundus Images: A Survey. In *IEEE Access* (Vol. 8, pp. 151133–151149). Institute of Electrical and Electronics Engineers Inc. <https://doi.org/10.1109/ACCESS.2020.3015258>
- Simul Hasan Talukder, M., & Author, C. (n.d.). *An Improved Model for Diabetic Retinopathy Detection by using Transfer Learning and Ensemble Learning*.
- Tsiknakis, N., Theodoropoulos, D., Manikis, G., Ktistakis, E., Boutsora, O., Berto, A., Scarpa, F., Scarpa, A., Fotiadis, D. I., & Marias, K. (2021). Deep learning for diabetic

retinopathy detection and classification based on fundus images: A review. In *Computers in Biology and Medicine* (Vol. 135). Elsevier Ltd.
<https://doi.org/10.1016/j.combiomed.2021.104599>

Tymchenko, B., Marchenko, P., & Spodarets, D. (2020). *Deep Learning Approach to Diabetic Retinopathy Detection*. <http://arxiv.org/abs/2003.02261>